

1. Executive Overview and Research Context

Through my independent research into large scale industrial manufacturing operations and my academic knowledge of **Quality Engineering** and **Total Quality Management**, I have formulated a robust supply chain reliability framework tailored for **JSW Steel** operations at **Vijayanagar** and **Dolvi**. Continuous steel production requires absolute synchronization across all material intakes. My proprietary analytical solution addresses chronic supply chain bottlenecks by integrating advanced reliability metrics and structured maintenance philosophies to eliminate external vulnerabilities.

2. Operational Problem Definition

Through secondary research and operational analysis of **JSW Steel** supply workflows, I identified two primary failure modes that jeopardize continuous mill schedules:

- **Late Deliveries of Raw Materials:** Critical manufacturing inputs such as iron ore coking coal and packaging supplies frequently suffer from severe transit and dispatch delays from external vendors, directly risking plant line starvation.
- **Inconsistent Material Quality:** Repeated chemical composition shifts and physical dimension variations in incoming material batches severely compromise downstream processing standards and finished product integrity.

3. Series Reliability in Supply Chain Flow

To eliminate these vulnerabilities, I have structured the entire material intake pipeline into a rigorous **Series Reliability** model. In this framework, every operational stage acts as a dependent link where failure at any single point disrupts the entire workflow:

- **STEP A: VENDOR SOURCING AND DISPATCH**
The initial control checkpoint where raw materials are prepared and dispatched by vendors. Enforcing strict dispatch compliance here establishes a high baseline reliability score.
- **STEP B: BULK TRANSPORTATION AND TRANSIT**
Logistics movement via rail and road corridors. Tracking transit velocity closely guarantees precise material arrival timelines at plant gates.
- **STEP C: PLANT GATE INSPECTION AND QUALITY SCREENING**
Mandatory quality screening upon arrival. Any detected chemical or physical variation results in immediate batch rejection, preventing defective inputs from entering shop floors.
- **STEP D: WAREHOUSE STORAGE AND SHOP FLOOR FEED**
Final secure routing into active production lines, ensuring uninterrupted material availability for heavy rolling mills.

4. Reliability Metrics: MTBF and MTTF Implementation

To quantify and control operational downtime across our supply network, I have deployed precise reliability metrics separating reparable and non reparable failure domains:

- **MTBF Mean Time Between Failures for Reparable Elements:** I apply this metric strictly to operational systems that can be restored or repaired, such as vendor dispatch software platforms, logistics transport trucks, and internal material handling machinery. Tracking **MTBF** allows me to measure the exact average operating time between breakdowns and optimize corrective maintenance schedules.
- **MTTF Mean Time to Failure for Non Reparable Elements:** I utilize this metric for single use or disposable supplies and unfixable raw material batches that must be entirely scrapped. Tracking **MTTF** measures the operational interval between these unfixable rejections, driving better vendor accountability.

5. Strategic Maintenance Framework

To govern plant assets and supply chain touchpoints, I have established a robust maintenance strategy balancing preventive, predictive, and analytical approaches:

Maintenance Strategy	Core Operational Description	My Proposed Implementation
Preventive Maintenance PM	Time based or usage based periodic servicing executed regardless of current machine or system condition.	Mandatory vendor audit schedules and routine logistics fleet maintenance to prevent sudden breakdowns.
Predictive Maintenance PdM	Condition based monitoring utilizing performance data trends and automated alerts to catch failures before occurrence.	Analyzing historical delivery logs to forecast vendor delays and proactively activate backup suppliers.
Reliability Centered Maintenance RCM	An analytical decision framework used to determine the optimal maintenance policy for each specific asset.	Balancing inspection checkpoints across critical transport and storage junctures to minimize total operational cost.

6. Applied TQM Quality Tools

In developing this independent research solution for **JSW Steel**, I have embedded the following core Total Quality Management methodologies:

- **Series Reliability Modeling:** To mathematically evaluate end to end supply chain vulnerability.
- **MTBF and MTTF Analysis:** To segregate reparable operational downtime from non reparable batch rejections.
- **Vendor Quality Audits & SLA Enforcement:** To eliminate raw material variation at source.
- **Hybrid Maintenance Framework (PM, PdM, RCM):** To ensure asset availability and zero line starvation.

Authoritative Research Statement: This case study represents my independent professional research and engineering solution developed by **Jagriti Singh** for **JSW Steel** supply chain operations at Vijayanagar and Dolvi. All strategies, reliability frameworks, and metric implementations detailed herein are authored exclusively based on advanced quality engineering principles and academic research.